RAMANJEET SAINI

rj.s.saini@gmail.com | +1-(856)-271-6052 | linkedin.com/in/ramanjeet-saini | github.com/ramanjeetsaini

AI ENGINEER | APPLIED AI/ML & GENERATIVE AI SYSTEMS

EDUCATION

The University of Texas at Austin May 2023
M.S., Information Technology & Management | Concentrations: Machine Learning, Data Science | GPA: 3.96 / 4.0

Pandit Deendayal Energy University May 2018
B.Tech., Mechanical Engineering

TECHNICAL SKILLS

AI Systems: LLM inference, multi-agent orchestration, RAG, hybrid retrieval, similarity search, model evaluation, guardrails, model serving | **GenAI:** LangChain, LangGraph, LlamaIndex, Claude (AWS Bedrock), GPT-4o, Llama 3, QLoRA, DPO, vLLM, RAGAS, DeepEval, LangSmith | **ML / Serving:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Triton Inference Server, ONNX, FAISS | **Cloud & MLOps:** AWS Bedrock, SageMaker, Lambda, API Gateway, MLflow, W&B, Airflow, Docker, Evidently AI | **Data:** OpenSearch, Redis, PostgreSQL, DynamoDB, AWS S3, Spark, Kafka, Kinesis | **Languages & APIs:** Python, SQL, FastAPI, REST, GitLab CI/CD

EXPERIENCE

Western Union *Austin, TX*
AI / Machine Learning Engineer Aug 2023 – Present

- Led the end-to-end **APN prefunding** project, developing country-specific time-series forecasting models to support prefunding decisions.
- Built a **DRT agentic workflow** to review high-risk customer details, coordinating **OpenSearch** checks, affordability assessment from financial documents, and name matching through specialized agents.
- Developed **payment-card scanning and PAN masking** to protect cardholder data and support **PCI compliance**.
- Built a **KYC government-ID extraction API** using **Claude on AWS Bedrock and Amazon Textract**, storing extracted identity data encrypted for fraud prevention; achieved **97% precision / 94% recall** at **220 ms p95** and saved **$0.9M/quarter** in manual-review costs; served via FastAPI / API Gateway / Lambda in support of GDPR / AML compliance.
- Improved model quality by **14 F1 points** by fine-tuning **Llama 3.1 8B with QLoRA (PEFT)** and aligning via **DPO (TRL)**; served with **vLLM and AWQ 4-bit quantization**, reducing GPU memory use by **55%** with <2% accuracy drop.
- Hosted **Gemma and Qwen** on bare-metal servers for an on-prem AI pod; prototyping fine-tuning and prompt-based adaptation using company-wide knowledge and code repositories.
- Cut fraud-analyst handle time by **26%**, reduced false-positive rate by **14%**, and raised straight-through processing by **19 pp** by productionizing a **RAG pipeline** over OpenSearch with hybrid dense/sparse retrieval using **LlamaIndex + Claude on AWS Bedrock**; evaluated with **RAGAS** and traced with **LangSmith**.
- Protected a **35 bps** margin target and lifted checkout conversion by **9%** by designing a production **multi-agent pricing platform** using **LangGraph + Claude 3 (AWS Bedrock)**; applied Bedrock Guardrails and AWS KMS controls and eliminated **80%** of manual analyst effort.
- Drove **$1.1M/quarter** incremental revenue by architecting a **two-tower retrieval and ranking model (PyTorch + FAISS)** deployed on **Triton Inference Server with ONNX** at <80 ms p95; validated via A/B experiment.
- Lifted customer retention by **12 pp** and cut model release cycle by **30%** by owning end-to-end MLOps for churn and CLV models on **SageMaker + Kubeflow Pipelines** with MLflow, W&B, and Evidently AI drift monitoring.

Infosys Limited *Pune, India*
Data Engineer Jul 2020 – Jun 2022

- Cut infrastructure costs by **34%** by managing **Apache Spark / AWS EMR** batch pipelines processing 500M+ records/week with ML-based anomaly detection and **AWS Kinesis** real-time streaming.
- Improved ML personalization pipeline throughput by **12%** via **PySpark + Kafka → S3 → DynamoDB** pipelines; cut serverless costs by **33%** by optimizing AWS Lambda inference functions.

Cloud Engineer Oct 2018 – Jul 2020

- Reduced DBaaS (PostgreSQL, SQL Server, Oracle) downtime by **28%** via CloudWatch monitoring; deployed Tableau connectors via GitLab CI/CD, improving data reliability by **18%**.

PROJECTS

Snap & Track: Photo-Based Meal Logging App 2025

- Built a meal-logging app that uses **GPT-4o Vision** to identify foods and estimate calories from a phone photo, making it easier for users to record meals and review nutrition estimates.

Flight Deal Finder: Cash vs. Miles Comparison App 2024

- Built a **Streamlit** travel app that helps users compare flight options by cash price and miles value; used **LangChain, GPT-4o, and FAISS** to find and rank relevant fare and award options.